

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EC-6004-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

VLSI Circuits and Systems

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) With neat sketches explain the architecture of FPGA.
b) Compare PROM, PLA and PAL.
2. a) Implement 2×1 MUX using 4 CMOS transistor.
b) Implement 3-inputs CMOS NAND gates and explain its operations.
3. a) MOS transistors could be worked as switch. Verify this statement with the help of suitable diagram.
b) How CMOS is different from NMOS and PMOS? Explain.
4. a) Define Pulse mode asynchronous sequential machine with example.
b) Explain Stuck at 1 and stuck at 0 conditions with examples.
5. a) Name different types of Faults. How to detect faults?
b) Design a Binary to Gray code converter using
 - i) PROM
 - ii) PLA
 - iii) PAL

EC-6004-CBGS

PTO

[2]

6. a) Explain in detail about VLSI design flow.
b) Using CMOS logic, design a circuit for the function

$$F = \overline{(A+B)(C+D)}.$$

7. Draw the ASM chart for
i) AND Gate
ii) SR flip-flop

OR

Implement the following Boolean functions using a suitable PAL

- i) $w(A, B, C, D) = \sum_m(1, 3, 4, 6, 9, 11, 12, 14)$
ii) $z(A, B, C, D) = \sum_m(2, 3, 8, 9, 12, 13)$

8. Write short notes on:
a) GAL
b) Fault diagnosis
c) CAD Tools
